Massachusetts General Hospital
Founding Member, Mass General Brigham

George A. "Doc" Lopez, MD
Laboratory for Regenerative Cell Therapy
White 536, 55 Fruit St, Boston, MA 02114

Process	**α-synuclein aggregation with preformed fibrils (PFFs) *in vitro***		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Media					
PFFs					
TX-100					
Tris-buffered saline					
Protease inhibitors					
Phosphatase inhibitors					
SDS (10%)					
DBPS					

Days 21–27:

Instruction	**Data**	**Scribe**
On D21, treat DAPs with 0.5 μM PFFs in 2 mL of media. Change media (with no PFFs added) on D23, D25, and D27.	D21 with PFFs: □ D23: □ D25: □ D27: □	

Harvest:

Instruction	**Data**	**Scribe**
Cool the centrifuge to 4°C in advance and prepare a bucket of (wet) ice.	□	
On D28, harvest via differential detergent extraction. First, prepare your reagents:	□ □	

Soluble fraction (1% TBS)		
TBS	10 mL	
TX-100	100 μL	
Protease inhibitor	1:1000	
Phosphatase inhibitor	1:100	
Total	10 mL	

Insoluble fraction (2% SDS)		
TBS	800 μL	
SDS (10%)	200 μL	
Protease inhibitor	1:1000	
Phosphatase inhibitor	1:100	
Total	1 mL	

Vortex both to mix. Keep the soluble fraction solution on ice and the insoluble fraction solution at room temperature (to prevent the precipitation of the SDS).

Prepare a 1.5-mL tube per well in the BSC to collect your cells after harvest.	□	
Remove your cells from the incubator. Aspirate the media and wash with DPBS.	□	

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Aspirate, then add another 1 mL of DPBS and use a cell scraper to harvest the cells. Collect each well in your tubes.	◻	
Centrifuge at 5.0 rcf for 3 minutes at 4°C.	◻ ___ rcf ___ minutes	
Carefully aspirate the supernatant. Resuspend the pellet with 200 µL of your soluble fraction solution per well, pipetting up and down gently to mix. Incubate on ice for 30 minutes.	◻ Time: _____	
Centrifuge at 13.0 rcf for 40 minutes at 4°C. During this time, prepare a second set of 1.5-mL tubes to collect the soluble fraction.	◻ ___ rcf ___ minutes	
Using a pipet, reserve the supernatant(s) in your second (set of) tube(s). This is the soluble fraction of your harvest.	◻	
Resuspend the pellet with 200 µL of your insoluble fraction solution per well, pipetting up and down gently to mix.	◻	
Sonicate 10 times at a 1-sec. pulse on 90% power to completely disperse the pellet.	◻ _____ power	
Centrifuge at 13.0 rcf for 40 minutes at 4°C. During this time, prepare a third set of 1.5-mL tubes to collect the soluble fraction.	◻ ___ rcf ___ minutes	
Using a pipet, reserve the supernatant(s) in your third (set of) tube(s). This is the insoluble fraction of your harvest.	◻	
NOTE: This harvest can be completed without the PFF seeding to look at total and phosphorylated αS as well.		

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**Cell harvest for Western blot**		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
NP-40 lysis buffer (BP-119)					
DPBS					
Protease inhibitor					
Phosphatase inhibitor					

Harvest:

Instruction	**Data**	**Scribe**
Cool the centrifuge to 4°C in advance and prepare a bucket of (wet) ice.	◻	
Prepare a 1.5-mL tube per well in the BSC to collect your cells after harvest.	◻	
Remove your cells from the incubator. Aspirate the media and wash with DPBS.	◻	
Aspirate, then add another 1 mL of DPBS and use a cell scraper to harvest the cells. Collect each well in your tubes.	◻	
Centrifuge at 5.0 rcf for 3 minutes at 4°C.	◻ ___ rcf ___ minutes	
During this time, prepare your lysis buffer, to be kept on ice:	◻	
Carefully aspirate the supernatant. Resuspend the pellet with 50 µL of lysis buffer, then incubate for 15 minutes on ice.	◻ Time: _____	
Centrifuge at max. speed for 30 minutes at 4°C.		
Using a pipet, reserve the supernatant as your lysate. Discard the pellets.		

Lysis buffer	Volume added:
NP-40 lysis buffer	
Protease inhibitor (1:1000)	
Phosphatase inhibitor (1:100)	
Total	

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**α-synuclein Western blot**		
Date & time		Operator	

Bill of materials:

Lysates		ID	
Harvest date		# samples	
Harvest type		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Formaldehyde					
LDS (5x)					
MES buffer					
Ladder					

Running the gel:

Instruction	**Data**	**Scribe**
Prepare enough 1.5-mL tubes to accommodate your samples.	◻	
Add 12 µL per sample and 3 µL 5x LDS per tube.	◻	
Boil samples for 4 minutes in the hot rack, then briefly spin down to collect and mix.	◻	
Prepare your 1x MES running buffer.	◻	
Load 3 µL of ladder into the first lane, then 15 µL of your samples per well.	◻	
Run at 50V to begin, then proceed to 80–85V.	◻	

Transferring and imaging:

Instruction	**Data**	**Scribe**
Transfer the gel.	◻	
Prepare 4% paraformaldehyde to fix the blot. Incubate for 4 minutes with agitation.	◻	
Block for 45 minutes in 5% milk TBS with agitation.	Time in: _____	
During this time, prepare your primary antibodies by diluting in 5% milk TBS at 1:1000. Incubate for 1 hour at room temperature or overnight at 4°C.	Temp.: _____ Time in: _____	
Wash for 10 minutes in TBS-T three times with agitation.	◻ ◻ ◻	
During this time, prepare your secondary antibodies by diluting in in 5% milk TBS at 1:1000. Incubate for 30 minutes at room temperature, then wash as before and image accordingly.	Temp.: _____ Time in: _____ ◻ ◻ ◻	

All steps completed? ◻

Scribe: _____
(Initial & date)

Notes:

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Process	**Lysosomal-autophagic flux**		
Date & time		Operator	

Bill of materials:

Cells	Midbrain dopaminergic progenitors	ID	
Passage #		Plate size	
Day		# plates (wells)	
Density		**Scribe**	

Reagent	Lot #	Quantity/unit	# units	Expiry	**Scribe**
Media					
Bafilomycin A1					
Rapamycin					
Anti-LC3 primary					
Anti-actin primary					
Anti-p62 primary					

Days 21:

Instruction	**Data**	**Scribe**
4 hours prior to harvest, treat the cells with autophagy factors to stimulate (rapamycin) or inhibit (bafilomycin A1) autophagy: First, prepare your factors by diluting in media: **Bafilomycin A1 media**: Media 10 mL ☐; Bafilomycin A1 200 nM ☐ **Rapamycin media**: Media 10 mL ☐; Rapamycin 20 nM ☐ Untreated wells should receive media with a comparable dose of DMSO.	☐	
Remove your cells from the incubator. Aspirate the media and add 1 mL treatment media per well. Return the cells to the incubator for 4 hours. Note which treatments are added to each well.	☐ Time in: _____	

Harvest:

Instruction	**Data**	**Scribe**
4 hours after treatment, harvest cells as described in the **cell harvest for Western blot** worksheet.	☐	
Blot for LC3, p62, and actin. Lysosomal flux can be measured by taking the difference between		

All steps completed? ☐

Scribe: _____
(Initial & date)

Notes: